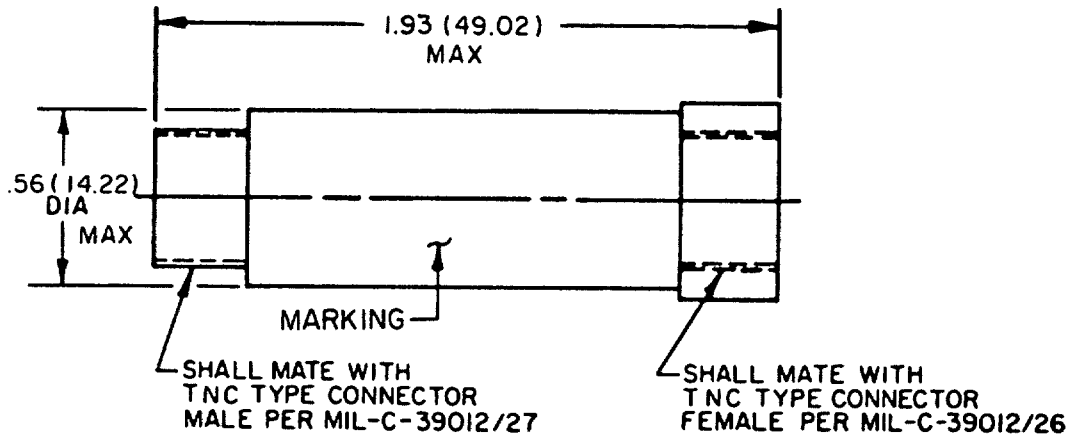
MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (TNC CONNECTORS),
FREQUENCY RANGE: DC TO 18 GHz, CLASS IV, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue of MIL-A-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Class: IV.

Operating frequency range: DC - 4.5 GHz for part numbers M3933/17-01 thru -06; DC - 18 GHz for part numbers M3933/17-07 thru -24.

Test frequencies for VSWR and attenuation measurements: DC, 2.25, and 4.5 GHz for part numbers M3933/17-01 thru -06; DC, 4, 8, 12.4, and 18 GHz for part numbers M3933/17-07 thru -24.

Basic test frequency: 2.25 GHz for part numbers M3933/17-01 thru -06; 8 GHz for part numbers M3933/17-07 thru -24.

Operating temperature range: -55°C to 125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I.

Power: See table I.

VSWR: See table I.

Part number: M3933/17- (and dash number from table I).

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB				Maximum power input @ 25°C		VSWR maximum 1/					Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation			AV (W) 3/ (continuously)	Peak 2/ (kW)	DC to 4.5 GHz	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
		DC	DC to 4.5 GHz	DC to 18 GHz									
01 N S	1	±0.15	±0.3	---	4	0.2	1.25:1	---	---	---	---	3.0	---
02 N S	2	±0.2	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
03 N S	3	±0.2	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
04 N S	4	±0.2	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
05 N S	5	±0.2	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
06 N S	6	±0.2	±0.4	---	4	0.2	1.25:1	---	---	---	---	3.0	---
07 N S	20	±0.4	---	±0.6	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
08 N S	30	±0.6	---	±0.9	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
09 N S	1	±0.2	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
10 N S	2	±0.2	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
11 N S	3	±0.2	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
12 N S	4	±0.2	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
13 N S	5	±0.2	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB				Maximum power input @ 25°C		VSWR maximum 1/					Weight (oz) Max	Dimension A (inches) maximum
	Nominal	Deviation					DC to 4.5 GHz	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz	12.4 to 18.0 GHz		
		DC	DC to 4.5 GHz	DC to 18 GHz	AV (W) 3/ (continuously)	Peak 2/ (kW)							
14 N S	6	±0.2	---	±0.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
15 N S	7	±0.3	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
16 N S	8	±0.3	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
17 N S	10	±0.3	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
18 N S	12	±0.3	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
19 N S	15	±0.3	---	±0.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
20 N S	25	±0.5	---	±0.8	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
21 N S	35	±0.6	---	±0.9	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
22 N S	40	±0.6	---	±1.0	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
23 N S	50	±0.8	---	±1.4	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---
24 N S	60	±0.9	---	±1.5	2	0.2	---	1.15:1	1.20:1	1.25:1	1.35:1	3.0	---

1/ VSWR value is for both ends.

2/ Peak power for a duty cycle of 5×10^{-4}, maximum pulse duration of 5 microseconds.

3/ Power input is derated linearity to 0.5 watts at +125°C.

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Note: Attenuators covered by this specification sheet are substitutable for the manufacturer's or user's part numbers specified in table II. This information in no way implies that manufacturer's part numbers is suitable as a substitute for the military part numbers.

TABLE II. Substitution information.

Military part number M3933/17-	Manufacturer's code	Manufacturer's part number	NSN
01	49956	2887496-1	---
01	23042	FP-50-1dB-TNC	---
01	00929	AA-01T	5985-00-989-9657
02	00929	AA-02T	5985-00-989-9658
02	00929	AB-02T	5985-00-784-8448
02	49956	2887496-2	---
02	23042	FP-50-2dB-TNC	---
03	00929	AB-03T	5985-00-138-7097
03	00929	AB-03T	5985-00-901-1248
03	00929	AA-03T	5985-00-959-2901
03	49956	2887496-3	---
03	23042	FP-50-3dB-TNC	---
04	49956	2887496-4	---
04	23042	FP-50-4dB-TNC	---
05	49956	2887496-5	---
05	23042	FP-50-5dB-TNC	---
06	00929	AB-06T	5985-00-985-2420
06	49956	2887496-6	---
06	23042	FP-50-6dB-TNC	---
07	49956	2887496-7	---
07	23042	FP-50-20dB-TNC	---
07	23042	FP50-20TNC	5985-00-534-7539
07	93459	18T20dB	5985-01-140-5610
07	00929	AH-20T	5985-00-496-9789
07	00929	AA-20T	5985-00-740-1307
07	00929	AB-20T	5985-00-917-4953
07	18203	A420T	5985-00-175-0247
07	18203	A8120T	5985-00-483-2126
08	23042	FP-50-30dB-TNC	---
08	00011	30-106613	---
08	18203	A430T	5985-00-175-0265
08	18203	A330T	5985-01-030-7820
08	99999	779T30	5985-00-390-8571
08	00929	AB-30T	5985-00-840-2814
09	18203	A401T	5985-00-131-6735

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TABLE II. Substitution information - Continued.

Military part number M3933/17-	Manufacturer's code	Manufacturer's part number	NSN
11	18203	A303T	5985-00-138-7097
11	18203	A403T	5985-00-618-9436
11	99899	757T3	5985-00-927-7670
11	00929	AH-03T	5985-00-624-4926
14	99899	757T6	5985-00-028-4142
14	18203	A406T	5985-00-130-6553
14	57958	5054269-2	---
14	93459	18T-6	---
16	00929	AA-08T	5985-01-070-3530
17	00929	AH-10T	5985-00-100-4723
17	00929	AB-10T	5985-00-138-7098
17	00929	AB-10T	5985-00-905-4017
17	00929	AA-10T	5985-00-959-2900
17	57958	505-4269-1	---
17	93459	18T-10	5985-01-152-5133
17	23042	FP50-10dBTNC	5985-01-029-0893
17	18302	A310T	5985-00-138-7098
17	18302	A410T	5985-00-175-0246
17	18302	A310T	5985-00-422-4978
18	00929	AA-12T	5985-00-247-0413
19	00929	AA-15T	5985-00-989-9659
19	18203	A415T	5985-01-075-8575
21	18203	A335T	5985-00-256-5022
21	00929	AB-35T	5985-00-256-5022
22	00929	AA-40T	5985-00-147-0717
24	00929	AA-60T	5985-01-021-7025
24	00724	10-00393	---

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC, SH, AS

Agent:
DLA - ES

Preparing activity:
Navy - EC

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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